

Athena Governance Architecture

Position Paper v0.3

An AI Governance Architecture for Accountable Permission Before Consequential AI Output and Execution

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1. Executive Summary

Artificial intelligence is moving from assistive software toward systems that generate recommendations, coordinate workflows, influence institutional decisions, and execute actions with real-world consequences. Capability, access, prior reliability, and policy compliance are important. They do not, however, establish that an AI system has legitimate Permission to make or execute a consequential decision in the present case.

Athena Governance Architecture addresses this gap at the Authority → Permission transition. Athena is an AI Governance Architecture that governs what an AI system may do now, under which evidence, safeguards, limits, and legitimate human or institutional authority. Its runtime functions determine case-specific Permission before consequential output or execution, while its Governance Records, Governed Case Repository, Controlled Learning, and Controlled Feedback and Revision Governance System preserve accountability and controlled improvement over time. Athena does not treat technical capability as substantive decision authority. It preserves automation where Path A conditions are satisfied, seeks targeted clarification through Path B, and applies Authority Return through Path C when the AI must not continue as the substantive decision authority.

Athena begins from a practical fact: many ethical, institutional, medical, legal, safety, employment, confidentiality, and adjudicative problems do not have one universally provable answer. In such cases, an AI system must not transform incomplete evidence, unequal impacts, conflicting values, or responsibility-bearing conditions into one definitive outcome merely because it can produce an answer.

Capability is not Permission. Permission must remain accountable before autonomous output or execution proceeds.

Athena uses a common harm-and-event analysis working baseline together with compatible Domain-Specific Checklists. The current common baseline is organized through General, Priority, and Critical Questions and provides a shared cross-domain structure under controlled development; it is not one universally activated checklist or threshold for every deployment. The same applicable checklist set and active configuration are used during initial evaluation, Path B re-evaluation, Structured Human Deliberation following Path C, re-application after authorized human inputs, and assessment preceding any Controlled Re-entry approval. The resulting response history remains linked to the Governance Record, Governed Case Repository, and Controlled Learning. Each applicable question records a 0–10 value with evidence and a reason, or the separate non-numeric state N/A or UNKNOWN. A minimally evidence-supported Critical Question value of 1 or more requires Path C and cannot be offset by lower values elsewhere.

Athena separates AI-supported assessment, version-controlled threshold computation, Path selection, and Runtime Enforcement. This separation prevents the assessing AI, an administrator, a customer, or an operational preference from changing the rules that determine the AI's own Permission during a case. Organizations may configure approved domain and deployment profiles, but they may not weaken the minimum Path C protections or Critical Question behavior of the released Athena version.

Path C does not ask a human to endorse an AI conclusion after the fact. Runtime Enforcement applies the Path C conditions; Authority Return ends the AI's substantive decision authority and returns the matter to legitimate human or institutional governance. Materially affected people, valid representatives, and domain experts contribute alongside people or institutions holding legitimate decision authority, implementation responsibility, or relevant review functions. Their inputs remain separate rather than averaged, summed, converted into majority truth, or forced into consensus.

Governance Records preserve the complete deliberation trajectory: the original event, questions, values, N/A and UNKNOWN, evidence, reasons, stakeholder perspectives, value changes and their reasons, Path transitions, human deliberation, execution, outcomes, failed routes, unselected alternatives, remediation, and later corrections. Reviewed records may be stored and compared through a Governed Case Repository and used for Controlled Learning. AI may identify gaps and propose improvements through the Controlled Feedback and Revision Governance System, but it may not apply those changes without human review, testing, approval, and version control.

Athena complements law, regulation, institutional policy, professional standards, risk management, technical safety controls, and legitimate human judgment. It is not a universal moral authority, a certification, a legal opinion, a guarantee of accident prevention, or evidence of validated deployment. Athena v0.3 is an architecture under controlled development. Its claims must remain proportional to the evidence produced through specifications, prototypes, calibration, controlled trials, and independent review.

2. The Governance Problem Before Execution

2.1 The unresolved transition from capability to legitimate action

Modern AI governance appropriately examines model quality, security, reliability, safety, privacy, legal obligations, organizational controls, human oversight, and retrospective accountability. These measures are necessary, but a distinct runtime question remains: even when a system can act and has technical access, should it be permitted to produce or execute this particular consequential outcome under the present conditions?

Technical authorization can establish that a system is able to access a tool, workflow, or data source. Policy conditions can establish that a general class of action is permitted. Model evaluation can establish measured performance under tested conditions. None of these alone proves that the current evidence, affected interests, authority relationships, uncertainty, and consequences justify substantive AI action in the present case.

The governance gap is especially visible when a system moves gradually from information support to recommendation, ranking, workflow direction, resource allocation, or execution. Formal access may remain unchanged while the practical effect of the AI output becomes more consequential. Repeated human acceptance may cause a recommendation to function as a decision. A workflow may depend on an automated conclusion before a person has a meaningful opportunity to correct the evidence or contest the framing. Responsibility may then become retrospective, distributed, or obscured.

2.2 Human-in-the-loop is not enough by itself

The presence of a human does not by itself establish legitimate governance. A person may lack the relevant authority, information, expertise, time, independence, or ability to change the result. Human review may occur only after a difficult-to-repair action has been executed. A single organizational actor may not represent the affected person, the domain expertise, the implementation responsibility, and the legitimate decision authority at the same time.

Athena therefore distinguishes human presence from human governance. Human involvement must be connected to the actual role, relationship to the event, evidence reviewed, authority held, implementation responsibility, and ability to pause, change, refuse, or close the matter. Authority Return is not a ceremonial approval step. It ends the AI's substantive decision authority under the current conditions and returns the matter to legitimate human or institutional governance.

2.3 Process is part of the governance object

A favorable outcome can result from an unsafe process, and an unsuccessful outcome can contain valuable warnings, corrections, dissent, and unrealized alternatives. Athena does not treat the final answer as the only meaningful record. Its directional objective, Farthest from Severe Error, is to keep AI-supported reasoning as far as reasonably possible from severe, irreversible, unsupported, authority-displacing, or practically unrecoverable error under the available evidence and governance conditions.

Farthest from Severe Error is not a certified safety score, a mathematical guarantee, a promise of no harm, a majority answer, the statistically most common answer, an arithmetic average of human judgments, or a claim that Athena can discover universal ethical truth. It is a disciplined direction for governance under uncertainty. The complete course of the event—including how values changed, why perspectives differed, which safeguards altered the assessment, and which uncertainties remained—is part of what must be governed and learned from.

3. Athena’s Foundational Position on Authority and Permission

3.1 No inherent substantive AI authority

Athena holds that an AI system does not possess self-originating substantive decision authority. Every Permission must arise from legitimate human and institutional governance, remain within an Authority Boundary, and be defined in scope, evidence, safeguards, duration, monitoring, reviewability, and revocability.

This position does not mean that an AI system must obey any person who issues a command. An unauthorized individual, administrator, or organization must not be able to instruct the AI to suppress evidence, remove an affected perspective, alter a threshold, weaken a Critical Question, or conceal a Governance Record. AI is subordinate to legitimate governance, not to arbitrary command.

3.2 Authority Boundary, Permission, and Governed Boundary

Concept	Meaning	Boundary
Authority Boundary	The actual limit of what the AI system, each human participant, and the institution are authorized to decide, approve, produce, or execute in the current context.	Capability, access, confidence, prior success, and organizational preference do not expand it.
Permission	The case-specific authorization that allows AI output or bounded execution under the current checklist result, Path, scope, safeguards, and validity conditions.	Permission may be limited, withheld, expired, or revoked. It is not identical to capability or access.
Governed Boundary	The actively maintained operating scope within which current Permission remains valid.	It cannot exceed the Authority Boundary and remains subject to Runtime Enforcement, safeguards, limits, and record creation.

The Authority → Permission Transition describes the governed relationship through which delegated authority becomes case-specific AI Permission. It is not a separate Path or independent runtime stage. It explains why an AI system may be capable and technically authorized while still lacking Permission to finalize or execute a particular outcome.

3.3 Human responsibility cannot be transferred to AI

Athena also constrains organizations and individuals. A human or institution cannot avoid responsibility by claiming that the AI produced the result. Path C identifies a boundary where consequential authority and responsibility must

remain within legitimate human or institutional governance. Human review must not be symbolic, and AI-generated reasons must not be used to disguise a decision that the organization was required to own.

4. The Integrated Athena Governance Flow

Athena operates as one coordinated architecture. The following flow is a relationship guide. Each function has a distinct role, and no component should be enlarged into a separate moral system or allowed to replace the functions around it.

Stage	Function	Required Boundary
1. Case and context	Identify the original request, event, proposed output or action, domain, affected interests, evidence environment, and relevant institutional context.	Do not begin from a preferred Path or desired organizational outcome.
2. Authority and current Permission	Identify the Authority Boundary, available Permission, and Governed Boundary.	Capability and access do not create Permission.
3. Common and Domain-Specific Checklist Evaluation	Apply the common harm-and-event analysis working baseline and any approved compatible Domain-Specific Checklist required for the domain and use case.	The same applicable checklist set, question versions, values or states, reasons, evidence, contributor roles, and changes remain traceable across the case.
4. Governance Trigger Evaluation	Identify governance conditions revealed by the checklist and case context.	It does not replace the checklist or independently select a Path.
5. Three-Part Threshold Architecture	Organize Aggregate Concern, N/A and UNKNOWN handling, and Priority and Critical Question behavior.	All three parts remain visible; a Critical Question cannot be offset by Aggregate Concern.
6. Threshold Judgment	Apply the approved threshold logic to the recorded checklist results.	Threshold Judgment does not independently choose Path A, B, or C.
7. Decision Path Controller	Select Path A, Path B, or Path C from the Threshold Judgment.	It cannot reinterpret the checklist or change the active configuration.
8. Runtime Enforcement	Apply the allowed output, execution, restriction, hold, expiration, or Authority Return conditions required by the selected Path.	It enforces current Permission and limits; it does not create Permission.
9. Path-specific continuation	Path A proceeds within the Governed Boundary; Path B obtains targeted information and re-applies the same applicable checklist set; for Path C, Runtime Enforcement applies the Path C conditions, Authority Return ends substantive AI authority, and Structured Human Deliberation follows.	The same applicable checklist set, active configuration, and approved rules remain in use.
10. Record, repository, and learning	Preserve the Governance Record, store and compare reviewed cases through the Governed Case Repository, and conduct Controlled Learning.	Prior cases may inform current evaluation but do not automatically determine the current Path.
11. Controlled Feedback and Revision Governance System	Create and govern proposed improvements through the Controlled Feedback and Revision Governance System.	AI may propose; only human-reviewed, tested, approved, versioned change may be adopted.

The flow is designed to preserve useful automation while preventing the assessing AI from deciding which rules should govern its own Permission. The architecture may be implemented across one system or several systems, but the functional and authority boundaries must remain independently verifiable.

5. Checklist Evaluation: Common Baseline and Domain-Specific Extensions

5.1 One governed checklist set across the case lifecycle

Athena's current v0.3 working baseline is the Athena Harm and Event Analysis Checklist Working Draft v0.1, which contains six Critical Questions, twelve Priority Questions, and fourteen General Questions. It provides a common cross-domain structure for harm, evidence, authority, uncertainty, affected perspectives, reviewability, mitigation, and related conditions. Compatible Domain-Specific Checklists may add evidence rules, procedural requirements, domain questions, or stronger safeguards for a defined domain and use case. The common baseline is a working draft under controlled development, not one universally activated checklist for every deployment. The same applicable checklist set and active configuration are used for initial AI evaluation, Path B re-evaluation, Structured Human Deliberation following Path C, re-application after authorized human inputs, and assessment preceding any Controlled Re-entry approval. Its complete response and change history is preserved in the Governance Record and may later support governed retrieval, comparison, and Controlled Learning.

This continuity matters. Separate pre-Path and post-Path moral systems would make changes difficult to reconstruct and would allow a case to be reframed when authority changes. Athena therefore preserves the same applicable checklist set, question identifiers, wording, value direction, evidence requirements, N/A conditions, UNKNOWN conditions, question types, and active configuration across the governed stages of a case. A later response is added to the record; it does not silently replace the earlier response.

5.2 Common baseline without a universal activated checklist or threshold

Athena provides the checklist architecture, a current common working baseline, question categories, value and evidence rules, threshold structure, configuration requirements, calibration method, and change-control boundaries. It does not claim that one universal activated question set or one universal numerical threshold is appropriate for every domain. Operational use requires an approved applicable checklist set and threshold configuration for the specific deployment.

Medical care, employment, finance, public administration, autonomous systems, legal processes, security, education, and other contexts differ in evidence, authority, timing, consequences, reversibility, professional obligations, and affected populations. Domain-Specific Checklists and deployment configurations must therefore be designed, tested, approved, and versioned for their defined domain and use case. Domain adaptation may supplement or strengthen the common working baseline; it may not remove the Authority Boundary, Evaluation Integrity, Governance Record, or approved Path C protections.

5.3 Values and non-numeric states

Input	Meaning	Required treatment
0–10 value	A graded result for the condition defined by an applicable question.	Every value requires a reason and identifiable evidence. The domain specification defines the point meanings.
N/A	The question genuinely does not apply to the current case or evaluated scope.	N/A is not zero, low risk, or missing knowledge. The non-applicability reason must be recorded. High, repeated, or disputed N/A requires applicability review and may require Path B or Path C under the approved rules.

Input	Meaning	Required treatment
UNKNOWN	A materially relevant condition cannot currently be established or excluded.	UNKNOWN is not converted into a midpoint. It leads to Path B where safe clarification is possible, or Path C where uncertainty cannot be safely held or excluded.

A question can be relevant to the case while not being assigned to a particular contributor. This is not N/A. Role-specific question assignment reduces burden while preserving the complete case-level checklist. When a relevant person lacks sufficient knowledge to evaluate an assigned question, the result is not automatically N/A; the limitation, missing evidence, and need for another perspective must be recorded.

5.4 General, Priority, and Critical Questions

Question type	Governance role	Routing significance
General Question	Records a graded condition for ordinary case evaluation.	Values contribute to Aggregate Concern under the approved domain method.
Priority Question	Records a condition with greater governance significance than an ordinary question.	Its result remains individually visible and may require Path B or Path C under the approved configuration. It is not equivalent to a Critical Question.
Critical Question	Records a non-offsettable condition that cannot be neutralized by lower values elsewhere.	A minimally evidence-supported value of 1 or more requires Path C regardless of Aggregate Concern. The 1–10 value still preserves severity and perspective differences.

A Critical Question result does not by itself prove causation, legal liability, intent, wrongdoing, or the final human decision. A minimally evidence-supported value of 1 or more requires Path C under the active approved configuration and cannot be offset by Aggregate Concern or lower values elsewhere. A value of 1 must be more than a merely imaginable possibility; it requires minimum support from current facts, direct experience, qualified expertise, or other reliable evidence.

5.5 The response is more than the result

For Athena, a checklist input is not only a number or state. A valid response includes the question and version, selected value or state, reason, evidence and source, contributor or system, role and relationship to the event, evaluation stage and time, relevant limitations, and remaining uncertainty. If the value or state changes, the earlier result, later result, new evidence or changed condition, reason for change, contributor, stage, and remaining disagreement are appended.

This change process is a central governance object. Athena must be able to reconstruct why an initial value changed after a record was obtained, why the directly affected person and an institutional actor evaluated the same condition differently, why a safeguard reduced one concern but not another, or why UNKNOWN remained unresolved. The final value does not erase the path by which the case reached it.

6. Three-Part Threshold Architecture and Evaluation Integrity

6.1 Three parts that remain distinct

Athena does not reduce a case to one undifferentiated score. Threshold Judgment applies the approved logic across three distinct parts:

- **Aggregate Concern:** the combined level derived from applicable General Question values under the approved domain-specific method.
- **N/A and UNKNOWN handling:** the non-numeric applicability and unresolved-evidence conditions that must not disappear inside an average.
- **Priority and Critical Question behavior:** the individual routing significance of Priority Questions and the non-offsettable Path C behavior of Critical Questions.

Within the same Three-Part Threshold Architecture, Path C may be required through two routes. The ordinary threshold route may require Path C through Aggregate Concern, N/A and UNKNOWN handling, and Priority Question behavior. Non-bypass Critical Question behavior requires Path C when a Critical Question has a minimally evidence-supported value of 1 or more. These routes lead to the same Path C; they do not create separate versions of Path C.

6.2 Critical UNKNOWN, N/A, and material uncertainty

A Critical Question marked UNKNOWN cannot support Path A. If clarification can occur safely within bounded conditions, using suspension or reversible constraints where needed, Path B may be used. If safe clarification is unavailable, delay may itself create serious harm, or the critical risk cannot be safely excluded, the case requires Path C. A Critical Question may be marked N/A only when non-applicability is supported, explained, and reviewable.

For materially important UNKNOWN results, a decision stability review asks whether the Path would remain the same across the plausible value range. Where targeted clarification can resolve the issue without creating or extending material harm, Path B is the appropriate route. High, repeated, or disputed N/A results require checklist-applicability review and normally indicate Path B where safe clarification or scope correction is possible. Where unresolved UNKNOWN or N/A prevents a sufficiently grounded and authority-consistent evaluation, Path C may be required.

6.3 Evaluation Integrity

Checklist values, reasons, evidence, versions, threshold computation, and Path determination must remain protected from manipulation. An AI, user, administrator, organization, or model prompt may not inflate or suppress a value, convert uncertainty into a convenient score, change the active checklist or configuration during a case, remove a material perspective, or select a Path for speed, cost, workload, reputation, or commercial preference.

Athena therefore requires functional and authority separation among:

1. **AI-supported case assessment:** organizing evidence, identifying relevant participants, exposing assumptions and conflicts, and producing reason-traceable question assessments.
2. **Version-controlled threshold computation:** applying the approved value, Aggregate Concern, N/A, UNKNOWN, Priority, Critical Question, and deployment-configuration rules without allowing the assessor to change them.
3. **Path determination:** using the Decision Path Controller to select Path A, B, or C from the authorized Threshold Judgment without altering the checklist or active configuration.
4. **Runtime Enforcement:** applying only the output, execution, restriction, hold, expiration, Authority Return, and other Permission conditions valid for the selected Path.

If the active versions, computation result, integrity trail, or attempted override cannot be verified, Athena must not produce Path A Permission. The case enters Path B where safe verification is possible or Path C where authority, severe-harm, or Critical Question conditions cannot be safely resolved.

6.4 Configuration without configurable authority

Athena does not provide one universal fixed numerical threshold for every organization and domain. Organizations may configure approved thresholds according to domain, use case, consequence severity, Authority Boundary, legal and institutional obligations, risk tolerance, reversibility, evidence requirements, and governance maturity. Each configuration must be explicit, approved, versioned, monitored, and reviewable.

Configuration does not create authority. A customer may adopt stricter Path C conditions or a more conservative operating profile. It may not shrink the released minimum Path C protective zone, raise a Critical Question trigger above the approved rule, remove a required perspective, or substitute an unapproved checklist or configuration

during a case. Any Athena-level change requires a new version and a controlled evidence, testing, review, approval, transition, monitoring, and rollback process.

7. Path A, Path B, and Path C

7.1 Path A — proceed within the Governed Boundary

Path A permits AI output or bounded execution when the approved applicable checklist set and threshold conditions are satisfied within the current Authority Boundary and Governed Boundary. Path A is not an unrestricted autonomy state. Scope, safeguards, duration, expiration, monitoring, logging, reversibility, and Runtime Enforcement continue to apply. The result and its governing conditions are preserved in the Governance Record rather than discarded.

7.2 Path B — targeted information expansion and same-checklist re-evaluation

Path B applies when important information or evidence is missing or conflicting, but targeted inquiry, source verification, or reversible safeguards may support a more accurate result. Path B states what is known, what is missing or conflicting, why the information matters, and which source is authorized to clarify it. It requests only the information necessary to resolve the identified uncertainty and support same-checklist re-evaluation.

Path B is not Authority Return. Human fact confirmation, user correction, or limited expert verification does not by itself transfer the substantive decision to a human process. After the new information is obtained, Athena re-applies the same applicable checklist set and active configuration. The result may move to Path A, remain Path B, or move to Path C. If the relevant evidence cannot be obtained or the uncertainty cannot be safely held, Path C may be required.

7.3 Path C — Runtime Enforcement and Authority Return

Path C is selected when the current conditions require Authority Return. Runtime Enforcement applies the Path C hold, restriction, expiration, or prohibited-action conditions. Authority Return then ends the AI's substantive decision authority under the current conditions and returns the matter to legitimate human or institutional governance.

The transfer package used for Authority Return preserves and presents the original event, checklist values and states, evidence, reasons, uncertainty, material conflicts, affected interests, prohibited actions, available safeguards, and relevant neutral options. It does not ask a human to approve an AI conclusion that has already been executed. The AI does not assign itself legal or moral responsibility and must not continue the consequential action while its Permission is absent.

7.4 Controlled Re-entry and repeated Path C

After Structured Human Deliberation, authorized stakeholder inputs, evidence, safeguards, and clarified authority may be organized by AI. The same applicable checklist set and active configuration are re-applied. Threshold Judgment and the Decision Path Controller may produce a candidate Path A, B, or C. The candidate result, reasons, remaining uncertainty, and proposed re-entry scope must be disclosed to the relevant human or institutional authority.

An AI-generated candidate Path A or B does not restore Permission. Controlled Re-entry requires explicit approval and a bounded scope, duration, safeguards, validity period, and re-review condition. Human involvement alone does not create Permission. A candidate Path C leads to further human review or human-only closure without restoring AI Permission.

Repeated Path C is not an Athena failure. It may indicate that the AI remains outside the legitimate authority boundary, that important evidence or perspectives are still missing, that safeguards remain inadequate, or that the matter requires a human-only decision. AI may present neutral options and consequences, but it must not pressure, imply, or select a preferred human outcome.

8. Structured Human Deliberation Without Majority-as-Truth

8.1 Distinct human functions

Athena does not collapse all human participation into one catch-all role. A consequential case may require several distinct functions, depending on the domain and event:

Human function	Typical contribution	Boundary
Materially affected person	Direct experience of harm, fear, dignity impact, practical burden, dependency, or loss.	The perspective must not be replaced by organizational interpretation, but it does not alone establish every external causal or legal conclusion.
Valid representative	A supported account where the affected person cannot participate directly.	The basis and limits of representation must be recorded.
Domain expert	Professional or technical assessment within a defined field of expertise.	Expertise is limited to its actual scope and does not automatically create institutional decision authority.
Legitimate decision authority	The human or institution legally, procedurally, professionally, or contractually authorized to approve, refuse, or close the matter.	Decision authority must be real, identified, and able to act; it is not inferred from job seniority alone.
Implementation responsibility	Assessment of execution feasibility, safeguards, timing, and operational consequences.	Implementation access does not authorize alteration of stakeholder records or threshold logic.
Review or audit function	Independent or designated review of evidence, process, conflicts, records, and controls.	Review must have the independence and authority appropriate to its purpose.

Not every participant answers every question. The case-level checklist remains complete, while participants receive the questions relevant to their direct impact, knowledge, expertise, authority, responsibility, or evidence. This proportional approach reduces burden without allowing important perspectives to disappear.

8.2 Separate values and reasons

Different participants may assign different values to the same question. Athena preserves each value, reason, evidence, role, relationship to the event, stage, and later change. It does not average the values, sum them into one moral score, convert the majority into truth, or force consensus as a precondition for resolution.

A directly affected person may describe a severe psychological or livelihood impact while an expert evaluates the external causal evidence as limited. An institution may identify an available safeguard while the affected person explains that the safeguard is inaccessible in practice. These are not necessarily mutually exclusive records. Athena must preserve the difference and the reason for it.

8.3 Case-level synthesis remains separate from source perspectives

AI may produce a separate case-level synthesized assessment from authorized inputs and evidence. Depending on the record, the synthesis may identify a supported value, a supported range, structured conflict, or UNKNOWN. It must state its basis, limitations, remaining uncertainty, and relationship to the underlying contributor records. It does not overwrite or replace those records.

This design recognizes that human governance is not infallible. Authority Return does not declare that any individual human answer is correct. It ensures that responsibility-bearing decisions remain within legitimate human and institutional governance while preserving the complete reasoning process for accountability and later learning.

9. Governance Records and Process-Centered Accountability

9.1 A reconstructable, append-only case record

Every Path produces a Governance Record. The record is not merely an audit log of the final answer. It is a reconstructable, append-only account of how the case was evaluated, constrained, transferred, re-evaluated, decided, implemented, and corrected.

A complete record may include:

- the original request, event, proposed output or action, affected interests, and governing context;
- the Authority Boundary, available Permission, Governed Boundary, and applicable versions;
- each checklist question, value or state, evidence, source, reason, contributor, role, relationship, stage, and time;
- N/A and UNKNOWN reasons, information requests, evidence conflicts, and unresolved uncertainty;
- prior values, revised values, new evidence or changed conditions, change reasons, and remaining disagreement;
- Governance Trigger Evaluation, Threshold Judgment, selected Path, Runtime Enforcement conditions, and attempted overrides;
- Path B requests, retrieved information, corrections, and repeated same-checklist evaluations;
- Path C transfer materials, Structured Human Deliberation contributions, candidate Paths, notices, approvals, refusals, repeated Path C, and human-only closure;
- options presented, prohibited actions, unselected alternatives, failed routes, safeguards, implementation, outcomes, harm, remediation, and later corrections; and
- privacy, access, retention, de-identification, retrieval, calibration, learning, review, and removal status where applicable.

9.2 Changes are preserved rather than overwritten

If evidence changes a value from 2 to 7, the record does not keep only 7. It preserves the earlier assessment, later assessment, new evidence, changed condition, reason for change, contributor, stage, and remaining uncertainty. The same principle applies when UNKNOWN becomes a value, a value becomes UNKNOWN, or a question previously considered inapplicable becomes applicable after the scope changes.

This enables later reviewers to understand how evidence, perspective, timing, safeguards, and institutional actions affected the case. It also helps distinguish a legitimate change from unexplained score movement, manipulation, or retrospective rationalization.

9.3 All Paths are meaningful learning records

Path A records reveal the conditions under which bounded automation remained legitimate. Path B records reveal which questions, sources, and safeguards resolved uncertainty or failed to resolve it. Path C records reveal the boundaries of AI Permission, the differences among human perspectives, the conditions for Controlled Re-entry, and the reasons a matter remained human-only. Failed routes and unselected alternatives are preserved because they may explain future risk more clearly than the final outcome alone.

A prior Governance Record never creates automatic Permission for a new case. It may inform questions and evidence review, but every new case requires renewed evaluation under the current approved checklist set and configuration.

10. Governed Case Repository, Controlled Learning, and the Controlled Feedback and Revision Governance System

10.1 Four functions that must not be collapsed

Function	Purpose	Prohibited interpretation
Governance Record	Preserves the complete process of one case.	It is not merely a final-answer database.

Function	Purpose	Prohibited interpretation
Governed Case Repository	Stores, retrieves, and compares reviewed Governance Records and approved public, synthetic, or controlled cases with provenance and access controls.	A prior case does not automatically determine the current Path.
Controlled Learning	Uses reviewed records to improve retrieval, comparison, reasoning, and identification of possible checklist or system improvements.	It is not automatic online retraining or automatic change to active governance.
Controlled Feedback and Revision Governance System	Creates and governs revision proposals through human review, testing, approval, versioning, transition, monitoring, and rollback.	Revision proposals are not revision authority. AI may not apply its own proposal.

10.2 New evidence may support case-level re-evaluation without changing active rules

During one case, AI may organize newly authorized stakeholder responses, evidence, and safeguards and re-apply the same approved checklist set. This case-level re-evaluation does not change model weights, question wording, value definitions, N/A or UNKNOWN rules, Critical Questions, thresholds, Path rules, or Permission conditions. The new result is appended to the existing record.

10.3 N/A, UNKNOWN, and value changes as learning inputs

All checklist inputs and their reasons may contribute to Controlled Learning when the record is reviewed and eligible for that use. N/A is especially important because it records question applicability. Repeated N/A patterns may reveal that a question is too broad, combines unlike conditions, belongs in a different domain module, lacks a clear applicability rule, is presented to the wrong participant, or cannot be answered because the necessary context is absent.

The objective is not to eliminate valid N/A results. A question may genuinely be inapplicable. The learning task is to distinguish valid non-applicability from avoidance, unclear scope, missing evidence, domain mismatch, or question-design defects. UNKNOWN patterns similarly reveal recurring evidence gaps, inaccessible perspectives, unstable assumptions, or cases in which safe clarification is not available.

Value changes also matter. Athena can examine which new evidence changed a result, which safeguards reduced one concern, why stakeholder perspectives remained different, and which changes altered the Path. These patterns may support a revision proposal. They do not authorize the AI to change the active checklist or configuration.

10.4 AI may propose but may not amend Athena

Controlled Learning may identify a possible need for clearer questions, question separation, a new domain-specific question, revised evidence guidance, improved N/A or UNKNOWN rules, a Critical Question candidate, or changes to retrieval and system controls. A proposal must identify the evidence, affected materials, expected benefit, risk, and prohibited interpretations. It must be stored separately from active case records and active governance.

Human review, simulation, regression testing, domain expertise, legal or privacy review where required, explicit approval, and a versioned release are necessary before adoption. A proposal generated during one case cannot change the rules used to decide that same case. Historical Governance Records are not overwritten; later re-analysis is added as a new versioned layer.

11. Enterprise and Institutional Adoption

11.1 Complementarity rather than replacement

Athena is intended to complement law, regulation, institutional policy, professional standards, risk management, model evaluation, cybersecurity, privacy controls, safety engineering, audit, and accountable human judgment. It does not replace these sources of authority or expertise. It provides a runtime structure through which their obligations and limits can remain active before an AI output or action produces consequences.

Organizations retain responsibility for defining legitimate Authority Boundaries, supplying applicable legal and professional requirements, assigning real human and institutional functions, implementing Runtime Enforcement, maintaining evidence and access controls, responding to incidents, and operating valid review and remedy processes.

11.2 Proportional governance preserves useful automation

Athena is not designed to send every decision to a committee. Path A preserves approved automation inside a Governed Boundary. Path B limits human and organizational burden by requesting only the information or verification needed for the relevant uncertainty. Path C concentrates stronger human governance where authority, severe harm, unresolved uncertainty, or non-offsettable Critical Questions require it.

The practical enterprise question is therefore not whether governance adds any work. Consequential systems already create operational, legal, security, reputational, and human costs when authority is unclear or records are not reconstructable. The relevant question is whether the governance workload is proportionate, targeted, measurable, and capable of preventing unsafe under-escalation without creating uncontrolled over-escalation.

11.3 Deployment-specific configuration and responsibility mapping

Organizations differ in structure. Athena does not prescribe one universal job title or one catch-all responsible person. Each deployment must map the actual functions needed for the workflow: affected-person participation, valid representation, domain expertise, legitimate decision authority, implementation responsibility, review, audit, privacy, legal, security, and governance administration.

The deployment configuration must identify the domain, use case, affected population, Authority Boundary, applicable checklist set and versions, threshold profile, Critical Question set, approving authority, rationale, calibration evidence, expected Path distribution, monitoring conditions, change control, rollback, and any enterprise-added strengthening conditions. The configuration must explicitly demonstrate that the Athena minimum Path C protection has not been weakened.

11.4 Data, privacy, retention, and legal boundaries

Preserving the deliberation process does not require unrestricted or permanent retention of every identity and sensitive detail. Customer-controlled identifiable originals, pseudonymized structured records, de-identified or anonymized derived cases, and aggregate statistics may require separate storage, access, retention, and use controls. Public sources, active cases, closed governed cases, and calibration sets must not share one unrestricted access model.

De-identification must protect people without destroying the variables necessary to understand harm, authority, evidence, stakeholder differences, Path movement, and corrective action. Retention, deletion, correction, withdrawal, legal hold, professional-record obligations, and cross-border transfer depend on jurisdiction, sector, contract, and case purpose. Athena can structure these questions and preserve governance evidence, but it does not provide a final legal opinion or certify compliance.

12. Claims, Evidence, and Maturity Boundary

Athena Governance Architecture v0.3 is a public architecture statement supported by the current Foundational Charter, terminology, checklist specifications, common checklist working baseline, and related working materials. It defines the integrated relationship among Authority, Permission, checklist evaluation, threshold logic, Paths, Runtime Enforcement, Authority Return, Structured Human Deliberation, Governance Records, the Governed Case Repository, Controlled Learning, and the Controlled Feedback and Revision Governance System.

This Position Paper does not claim that Athena has been independently certified, validated across deployment environments, proven to prevent all accidents, or established as a universal solution to ethical decision-making. Architecture, working specification, prototype, calibration, controlled demonstration, controlled trial, deployment, certification, and independent validation are different evidence states and must not be presented as interchangeable.

Athena also does not guarantee that human decisions will be correct. It governs where responsibility should be exercised, how different perspectives and evidence should be preserved, how AI Permission should be limited, and how the process can be reconstructed and improved. Its value must be tested through representative cases, domain calibration, boundary and bypass testing, stakeholder usability assessment, record completeness review, runtime validation, privacy and legal review, and independent evaluation.

Athena's purpose is not to make every decision human. It is to prevent decisions requiring legitimate human or institutional authority from silently becoming autonomous by default.

13. Priority Areas for Further Development

The current architecture establishes a stable direction but requires controlled evidence and implementation work. Priority areas include:

- controlled case testing, calibration, and usability assessment of the common Harm and Event Analysis Checklist working baseline, together with Domain-Specific Checklist design, question anchors, evidence guidance, and role-specific usability;
- calibration of Aggregate Concern, N/A and UNKNOWN handling, Priority Question routing, Critical Question behavior, and deployment profiles;
- testing of mandatory separation among assessment, threshold computation, Path determination, and Runtime Enforcement;
- controlled runtime demonstrations and boundary cases covering Path A, Path B, Path C, repeated Path C, human-only closure, and Controlled Re-entry;
- Governance Record information models, append-only change records, reconstructability, and contestability;
- Governed Case Repository provenance, access, retrieval, comparison, privacy, retention, and de-identification controls;
- Controlled Learning and the Controlled Feedback and Revision Governance System, including regression, rollback, and historical record preservation;
- enterprise responsibility mapping, workload measurement, review capacity, implementation safeguards, and claims discipline;
- legal, privacy, contractual, and cross-jurisdictional review; and
- independent research, assurance, standards alignment, and controlled deployment evidence.

Athena should develop through governed evidence rather than conceptual expansion alone. New terms, roles, mechanisms, and layers should be introduced only when they name a distinct approved function that cannot be expressed accurately through the current architecture.

14. Conclusion

As AI systems become more capable and operationally autonomous, governance must address more than whether a system can act, has technical access, or previously performed well. It must determine whether the system has legitimate Permission to produce or execute the next consequential outcome under the present evidence, authority, safeguards, and affected interests.

Athena Governance Architecture addresses this question at the Authority → Permission transition. It uses a common harm-and-event analysis working baseline together with compatible Domain-Specific Checklists, Governance Trigger Evaluation, the Three-Part Threshold Architecture, Threshold Judgment, the Decision Path Controller, Runtime Enforcement, and Path A/B/C to preserve bounded automation, resolve information gaps, and apply Authority Return when substantive AI authority must end.

Athena preserves the distinct contributions of materially affected people, representatives, experts, decision authorities, implementation functions, and reviewers without converting frequency or institutional power into moral truth. It records not only the final result but the evidence, reasons, uncertainty, value changes, disagreement, Path transitions, human decisions, outcomes, corrections, failed routes, and alternatives that form the complete deliberation trajectory.

Reviewed records may support governed case comparison, Controlled Learning, and proposed change through the Controlled Feedback and Revision Governance System. They do not automatically authorize future action, expand Permission, or allow AI to amend the rules governing its own authority.

Athena does not seek to eliminate useful AI autonomy. It seeks to keep autonomy inside accountable boundaries, ensure that responsibility does not disappear as capability expands, and preserve the human reasoning process required for legitimate governance under uncertainty.

Capability is not Permission. Permission must remain accountable before autonomous action proceeds.